Exchange Concepts Knowledge Check — Variant 08

Purpose: verify that the student has read How a Financial Exchange Works and internalized the core principles of a modern exchange, across its full scope — market history, order mechanics, risk and compliance, and technology infrastructure.

Name/ID

Instructions

- There are 30 questions.
- Each question has five statements (A-E) which may be false or true.
- Select all statements you believe to be correct by clearly filling the square before the question.
- Each question may have multiple correct statements (for some, all five statements may be correct).
- There are always at least one correct statement.

Scoring

- +1 point for each correct option selected.
- -2 points for each incorrect option selected.
- 0 points for options not selected.
- Final test score is capped at a minimum of 0: if raw total is negative, recorded score is 0.
- A passing score is 70% of the maximum rounded down to the nearest integer.

Questions

1. Capital raising and the exchange's role

- ▢ A. A company that self-funds via retained earnings avoids outside obligations but grows more slowly
- ▢ B. The secondary market is where a company receives proceeds each time its shares trade
- ▢ C. Underwriters conduct a roadshow to gauge institutional demand before setting the IPO price
- ▢ D. A liquid secondary market is described as a precondition that makes primary-market investment attractive
- ▢ E. Bonds, unlike shares, never trade on secondary markets once issued

2. Market history and language

- ▢ A. The NYSE specialist's paper ledger is the direct ancestor of "the book" in modern systems
- ▢ B. "Blue chip" is traced to a 1923 Dow Jones description of stocks trading at $200 or more
- ▢ C. "Going short" is traced to a merchant having more inventory than needed to cover a delivery
- ▢ D. Black Monday, 19 October 1987, saw markets fall roughly 22.6% in one day
- ▢ E. The Brady Commission's report followed, rather than preceded, the introduction of NYSE's first circuit breakers

3. Participants, once more

- ▢ A. A taker is the participant whose incoming order immediately executes against a resting order
- ▢ B. NYSE's term for its market makers is "Designated Market Makers"
- ▢ C. Demutualisation is described as converting for-profit exchanges into member-owned mutuals
- ▢ D. A quote, as distinct from an order, is a two-sided bid/ask instruction from a market maker
- ▢ E. Institutional investors are described as typically trading small, randomly timed orders like retail investors

4. Real-world exchange facts, once more

- ▢ A. CME Group is described as one of the world's largest futures exchanges
- ▢ B. Cboe is credited with pioneering the listed options market in 1973
- ▢ C. HKEX and SGX are both described as real-world exchanges relevant to developers
- ▢ D. NYSE's parent company, ICE, is itself listed on NYSE
- ▢ E. Every major exchange remains a non-profit, member-owned mutual today

5. Listing, delisting, and alternatives

- ▢ A. A stock closing below $1.00 for 30 consecutive days can trigger a deficiency notice with a cure period
- ▢ B. A direct listing establishes its opening trade through the exchange's normal opening auction mechanism
- ▢ C. Spotify's 2018 NYSE listing is cited as a pioneering direct listing
- ▢ D. A SPAC merger is described as requiring the target company to complete its own traditional IPO first
- ▢ E. Voluntary delisting can occur when a company is taken private

6. Order types, comprehensive review

- ☐ A. A stop-limit order trades price certainty for potential non-execution if the market gaps through the limit
- ☐ B. A trailing stop only ever moves in the direction favourable to the position, freezing otherwise
- ☐ C. An iceberg's replenishment typically retains its original queue priority rather than going to the back
- ☐ D. A midpoint peg order can offer price improvement to both sides of a trade simultaneously
- ☐ E. A combo order's legs are matched entirely independently, with no shared execution intent

7. TIF, a comprehensive review

- ☐ A. DAY orders are cancelled automatically at session end if unfilled
- ☐ B. GTC orders must be persisted and reloaded across sessions
- ☐ C. IOC orders can rest in the book awaiting a better price
- ☐ D. FOK requires a full, immediate fill or the entire order is cancelled
- ☐ E. ATC orders specifically target the price set by the closing auction uncross

8. Depth, spread, and market impact, a fresh scenario

- ☐ A. Given asks of $50.00/3,000, $50.05/2,000, $50.10/1,000, a 4,000-share buy sweeps the first two levels fully and part of the third
- ☐ B. The VWAP of that 4,000-share sweep is (3000×50.00 + 1000×50.05)/4000
- ☐ C. Market impact equals the sweep's VWAP minus the initial best ask
- ☐ D. A bid-ask imbalance of exactly 0.5 indicates a perfectly balanced book
- ☐ E. Iceberg reserves have no effect on the relationship between displayed and true available depth

9. Price-time priority, one more pass

- ☐ A. A price amendment always assigns a new timestamp, regardless of direction
- ☐ B. A quantity-decrease amendment is the one case that preserves the original timestamp
- ☐ C. CME uses FIFO for its equity index futures products
- ☐ D. Pro-rata allocation splits an incoming fill proportionally to resting order size at a price level
- ☐ E. Cancel-and-replace can, in some systems, retain the original order's priority if requested

10. Tick sizes, one more pass

- ☐ A. Storing prices as integer tick counts avoids the binary floating-point representation problem
- ☐ B. A price submitted off the valid tick grid is rejected before reaching the matching engine
- ☐ C. Rule 612 prohibits sub-penny quoting in NMS stocks priced at or above $1.00
- ☐ D. Tick size is treated as reference data that can vary by price band, product, and venue
- ☐ E. The Tick Size Pilot Program ran for a single trading day in 2016

11. The matching engine, a comprehensive pass

- ☐ A. Single-threaded processing is chosen to guarantee deterministic, auditable outcomes
- ☐ B. Sweeping a large order through multiple price levels can generate slippage
- ☐ C. Calendar spreads require the engine to evaluate both legs of the trade together
- ☐ D. Each symbol's order book is entirely independent of every other symbol's book
- ☐ E. A market order that cannot fill immediately rests quietly in the book

12. Market makers, one more pass

- ☐ A. A market maker earns the spread by buying at the bid and selling at the ask
- ☐ B. A maximum-spread obligation could prevent quoting a bid and ask more than a specified number of ticks apart
- ☐ C. Cancelling both legs of a quote immediately upon any fill is the described "Eurex-style inactivation" policy
- ☐ D. MMP protection periods are described as giving the market maker time to reassess before re-quoting
- ☐ E. A market maker's obligations exist without any corresponding privileges from the exchange

13. Auctions, one more worked scenario

- ☐ A. Given buy orders 1,000@$25 and 400@$26, and sell orders 900@$25 and 300@$24, the executable volume at $25 is min(1400, 900+300)=1,200
- ☐ B. In that scenario, the executable volume at $26 is min(400, 900+300)=400
- ☐ C. Comparing $25 (1,200 executable) and $26 (400 executable), $25 is the equilibrium price
- ☐ D. A seller who asked $24 in that scenario receives $25 if matched, one dollar better than their ask
- ☐ E. The equilibrium price is chosen without regard to how much volume would execute at each candidate

14. Trading sessions, one more pass

- ☐ A. Pre-open orders accumulate without matching until the opening auction runs
- ☐ B. Intraday auctions can serve illiquid instruments or act as a softer alternative to a full halt
- ☐ C. GTC orders reloaded at start-of-day are re-acknowledged to their originating participants
- ☐ D. The session state machine permits transitions not explicitly defined, provided they are logically sound
- ☐ E. Book state saves provide the starting point for a warm restart

15. Indexes, one more pass

- ☐ A. A stock split, absorbed via divisor adjustment, should not by itself change the index level
- ☐ B. Being removed from a major index tends to cause forced selling by index funds
- ☐ C. The S&P 500's committee-based approach contrasts with mechanically reconstituted families like Russell or FTSE
- ☐ D. Index-linked circuit breakers are described as applying to individual stocks rather than the whole market
- ☐ E. Free-float weighting excludes shares held by insiders or governments that do not trade

16. Pre-trade risk controls, one more pass

- ☐ A. A notional value check multiplies order quantity by price to catch decimal-point errors
- ☐ B. Credit limits require tracking outstanding order commitments as well as settled positions
- ☐ C. Rate limiting is checked using an in-memory, per-gateway counter
- ☐ D. Short sale flag validation is unrelated to whether a valid locate exists
- ☐ E. The cheapest checks are deliberately run last to save computation for accepted orders

17. Circuit breakers and collars, one more pass

- ☐ A. The US uses three explicit market-wide tiers: 7%, 13%, and 20%
- ☐ B. LULD varies the trigger band by tier while using a fixed halt duration
- ☐ C. A dynamic price collar compares an order's price to the previous session's official close
- ☐ D. China's 2016 mechanism was withdrawn after only four trading days
- ☐ E. Japan's approach continues trading at the daily limit rather than halting outright

18. SMP, kill switch, and mass cancel, one more pass

- ☐ A. Cancel Aggressor is described as the most common default SMP policy
- ☐ B. A kill switch can be triggered automatically upon a lost gateway connection
- ☐ C. A mass cancel requires reconnection and re-authentication before the participant can resume trading
- ☐ D. After a kill switch fires, a participant's connection is typically marked inactive
- ☐ E. Wash trading is illegal under most exchange manipulation regulations

19. Clearing, settlement, and margin, one more pass

- ☐ A. Novation makes the CCP the counterparty to both the original buyer and seller
- ☐ B. Variation margin realises gains and losses in cash on a daily basis rather than letting them accumulate silently
- ☐ C. DvP is designed specifically to prevent the kind of failure that produced Herstatt risk
- ☐ D. Maintenance margin breaches trigger a margin call demanding additional collateral
- ☐ E. US equities currently settle T+3

20. Trade busting and clearly erroneous trades, one more pass

- ☐ A. The clearly erroneous threshold table widens as the reference price falls into lower bands
- ☐ B. Review is applied to an entire order as a single unit rather than execution by execution
- ☐ C. A participant typically has a short filing window, commonly around 30 minutes
- ☐ D. Busting reverses positions and cash movements as though the trade never happened
- ☐ E. Price adjustment can preserve that a trade occurred while correcting an erroneous print

21. Regulatory surveillance, one more pass

- ☐ A. Spoofing involves cancelling orders before they can fill, once their market-moving effect is achieved
- ☐ B. The CAT database has been operational in the US since 2020
- ☐ C. A Suspicious Transaction Report is filed only when a regulator specifically requests one
- ☐ D. Layering uses multiple orders at various price levels to create a false impression of depth
- ☐ E. The audit trail is required to be complete, immutable, and retained for a defined regulatory period

22. Knight Capital, comprehensive review

- ☐ A. The incident occurred on a single misconfigured server out of eight deployed
- ☐ B. Approximately $440 million was lost in roughly 45 minutes
- ☐ C. Knight later merged with Getco LLC to form KCG Holdings
- ☐ D. Every one of Knight's orders was individually valid at the prevailing market price
- ☐ E. The incident is described as having no lasting influence on subsequent risk-control regulation

23. Technology architecture, comprehensive pass

- ☐ A. FIX is text-based; ITCH and OUCH are binary protocols developed by NASDAQ
- ☐ B. Market data commonly travels via UDP multicast while order entry commonly travels via TCP
- ☐ C. The gateway typically performs authentication and message-format translation
- ☐ D. The matching engine and gateway are described as sharing an identical release cadence
- ☐ E. CME's MDP3 is based on the SBE (Simple Binary Encoding) standard

24. Onboarding, resilience, and load balancing

- ☐ A. Conformance testing verifies a participant's system handles rejections, partial fills, and reconnection correctly
- ☐ B. Partitioning by symbol allows different matching engine instances to scale independently
- ☐ C. A single central order-ID counter is described as free of single-point-of-failure risk
- ☐ D. Monotonic timestamp enforcement helps prevent a clock jump from assigning a "past" timestamp to a new order
- ☐ E. The sequencing infrastructure of the input queue, not the wall-clock timestamp, is described as what ultimately determines match priority

25. Market data architecture and economics, comprehensive pass

- ☐ A. A snapshot gives a complete current view of the book; incremental updates describe only changes
- ☐ B. Conflation is acceptable for display purposes but not for systems needing complete tick data
- ☐ C. A proprietary direct feed generally offers lower latency than the consolidated SIP feed
- ☐ D. The SEC's Market Data Infrastructure rulemaking sought to expand SIP content and restructure revenue governance
- ☐ E. Exchanges are described as having no financial interest in how competitive the public SIP is relative to their own proprietary feeds

26. Latency, co-location, and observability

- [ ] A. Co-located servers can reduce round-trip latency to low single-digit microseconds on optimised networks
- [ ] B. PTP achieves sub-microsecond clock synchronisation accuracy
- [ ] C. Symptom-based alerts are generally favoured over cause-based alerts to reduce alert fatigue
- [ ] D. A blameless postmortem is intended to surface causes without individual punishment
- [ ] E. Deterministic worst-case latency is described as unimportant compared to average latency

27. Determinism, persistence, and reference data

- [ ] A. Determinism requires eliminating hidden randomness and mid-execution clock reads from the matching path
- [ ] B. A write-ahead log is the source of truth, with in-memory book state treated as derived
- [ ] C. A warm restart from a recent snapshot is typically faster than a full cold start
- [ ] D. An ISIN identifies a security independent of which specific exchange lists it
- [ ] E. Reference data is generally described as safe to leave unversioned since it changes so rarely

28. Corporate actions and options, comprehensive pass

- [ ] A. A stock split requires proportional adjustment of resting limit order price and quantity
- [ ] B. A reverse split can serve as a remedy for a minimum-bid-price listing deficiency
- [ ] C. American-style options can be exercised at any point up to expiry
- [ ] D. Cash settlement is standard for broad index options like SPX rather than physical delivery
- [ ] E. Pin risk arises only for options that are unambiguously deep in-the-money at expiry

29. Cryptocurrency and digital assets, comprehensive pass

- [ ] A. Core order book and price-time priority concepts are described as carrying over to crypto exchanges
- [ ] B. Most crypto markets operate 24/7 without a defined daily open or close
- [ ] C. A decentralised exchange's AMM replaces order matching with algorithmic pricing from a liquidity pool
- [ ] D. FTX is presented as an example of institutional separation working correctly to protect customers
- [ ] E. Precision requirements for assets like Bitcoin and Ethereum tokens are described as more demanding than traditional equities

30. Final cross-cutting numeric review

- [ ] A. VWAP for a position of 100 shares at $150 and 50 shares at $160 is $153.33
- [ ] B. A market maker's loss is $200 if a 400-share position bought at $150.30 is later marked at $149.80
- [ ] C. An index with $7,007,100,000,000 aggregate market cap launching at base value 1000 has a divisor near 7.0071 billion
- [ ] D. Knight Capital's approximately 45-minute kill-switch delay is cited as a central failure in that incident
- [ ] E. A 60% deviation from a pre-crash reference price was used as the bust threshold following the 2010 Flash Crash

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